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Symbolically Regressing Fish Biomass Spectral Data: A Linear Genetic Programming Method With Tunable Primitives

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ABSTRACT

Machine learning techniques play an important role in analyzing spectral data. Fish biomass spectral data is useful in fish production as it carries key chemical properties of fish meat. However, it is challenging for existing machine learning techniques to comprehensively discover hidden patterns from fish biomass spectral data because the spectral data often have considerable noise, while the training data are quite limited. To better analyze fish biomass spectral data, we modeled it as a symbolic regression problem and solved it by a linear genetic programming method with newly proposed tunable primitives. In the symbolic regression problem, linear genetic programming automatically synthesized regression models based on the given primitives and training data. The tunable primitives further improved the approximation ability of the regression models by tuning their inherent coefficients. Our empirical results from over ten fish biomass targets showed that our proposed method using tunable primitives improved the overall performance of fish biomass composition prediction. The synthesized regression models are compact and have good interpretability, allowing us to highlight useful features over the spectrum. Our further investigation verified the good generality of the proposed method across various spectral data treatments and other symbolic regression problems.

1 | Introduction

Aquatic food is a key contributor to global food security and livelihoods. The global production of aquatic animals reached about 185 million tonnes in 2022, with an annual growth rate of about 3.2% and an approximate total first sale value of USD 452 billion (FAO 2024). Aquatic food is an essential source of a wide range of nutrients, particularly minerals and fatty acids.

This paper aims to predict fish biomass compositions by spectroscopic techniques, such as infrared and Raman spectroscopy. This is an important step in aquatic food production lines since it estimates nutrition compositions efficiently without destruction

(Xue et al. 2022; Song et al. 2024; Gade et al. 2024). Spectroscopic techniques usually work with machine learning methods, such as partial least square regression (PLSR) (Usman and Begum 2023) and convolution neural networks (Huang et al. 2023a), to predict biomass compositions.

However, analyzing fish biomass composition via spectral data is vastly different from many existing machine learning benchmarks owing to its unique challenges (Zhou et al. 2024; Wood et al. 2022). There are three main challenges. First, the data quality and quantity are very limited. Fish spectral data often have considerable noises since we can only scan the fish meat by laser for a short time (e.g., less than 4 min) to avoid heating

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it (Herrero 2008; Qu et al. 2022; Ahmmed et al. 2023). The amount of data is insufficient because measuring the accurate modules of materials is costly and cumbersome. These likely result in overfitting in existing machine learning methods. Second, some existing machine learning models lack interpretability to assist human scientists in understanding spectroscopic data. Because many machine learning models are black-box models and might include millions of parameters, it is non-trivial to understand and reason about their predictions. However, interpretability is important in aquatic food production. Food industries prefer ensuring the correctness of decisions and selectively setting up their equipment based on the necessary wavelength, which reduces their cost. Last but not least, there is a high correlation among adjacent input features in spectroscopy data. Existing machine learning models hardly make use of this correlation to enhance performance.

In response to these outlined challenges, this paper proposes a linear genetic programming method with tunable primitives (denoted as LGP-TP) to automatically synthesize regression models for analyzing spectral data, i.e., regarding the spectral data analysis as a symbolic regression task. Linear genetic programming (LGP) has shown promising learning ability and has been successfully applied to symbolic regression problems (Brameier and Banzhaf 2007; Huang et al. 2022), classification (Fogelberg 2005; Sotito et al. 2022), and combinatorial optimization (Huang et al. 2024). Compared with other machine learning methods, LGP can automatically find concise models for describing knowledge in the given data (Banzhaf et al. 2024). The concise models produced by LGP help improve the generality on the limited and noisy training data regarding the Occam razor principle and the minimum description length principle (Barron et al. 1998). LGP outputs regression models with good interpretability (Mei et al. 2022), highlighting interesting regions of spectroscopic data. It helps human scientists understand which features (or peaks in the spectrums) are useful for prediction. In addition, LGP programs naturally reuse sub-programs (by calling registers) to design compact solutions and have a similar form to human-written programs (instruction-by-instruction representation). This allows LGP to develop sophisticated programs, such as harnessing control flow operators (e.g., IF and FOR-loop) (Brameier and Banzhaf 2007; Huang et al. 2024) and having multiple outputs (Brameier and Banzhaf 2007; Huang et al. 2023b).

However, existing LGP methods usually synthesize regression models based on predefined primitives (e.g., $+$, $\times$, and x_1).

The fixed primitives might limit the approximation ability of LGP (e.g., representing $\sin(3x + \pi)$ by given $\sin(\cdot)$ only is not straightforward). Existing studies have proposed some methods to generate the constant coefficients, such as ephemeral random constants (Koza 1992; Ferreira 2003), geometric semantic genetic programming (GP) with linear scaling (Nadizar et al. 2024), and GP with mixed integer linear programming (Huynh et al. 2018). However, these methods are either too random or cannot effectively handle the coefficients inside primitives.

To enhance LGP performance in spectral data analysis, we propose three kinds of tunable primitives that perform residual error approximation in LGP programs. The tunable primitives can represent a wide range of function mappings by tuning their parameters. Specifically, the three kinds of tunable primitives are 1) tunable terminals that accept input features and approximate the target outputs; 2) tunable functions that accept intermediate results of LGP programs and approximate the target outputs; and 3) a multivariate linear regression function that approximates the final values of multiple registers to the target outputs. This work is inspired by a theoretical conclusion that reducing the difference between the inputs and outputs helps improve the average performance of LGP (Huang et al. 2025). In other words, the proposed tunable primitives are essentially approximating the residual error between inputs and outputs over program execution.

To apply LGP-TP to analyze fish biomass spectral data, we first apply spectroscopic and chemical analysis techniques to obtain the input and ground truth target data of fish samples, respectively (Figure 1). The spectroscopic data is pre-processed to enhance data quality. Specifically, the spectroscopic data is normalized by standard normal variate transformation (SNV) and augmented by linear interpolation. Then, we train LGP-TP to synthesize regression models to predict the target data based on the inputs. After training, LGP-TP outputs the best regression models. The regression model predicts the biomass compositions of newly harvested fish samples on production lines based on their spectral data. This enables a fast and cheap estimation of biomass composition without expensive chemical analysis.

Our results show that LGP-TP achieves the best overall predicting performance over ten fish biomass targets, compared with six popular regression methods (including the commonly used regression method for spectral data, PLSR, and one state-of-the-art method tailored toward fish biomass prediction). Our statistical analysis confirms that LGP-TP has significantly better

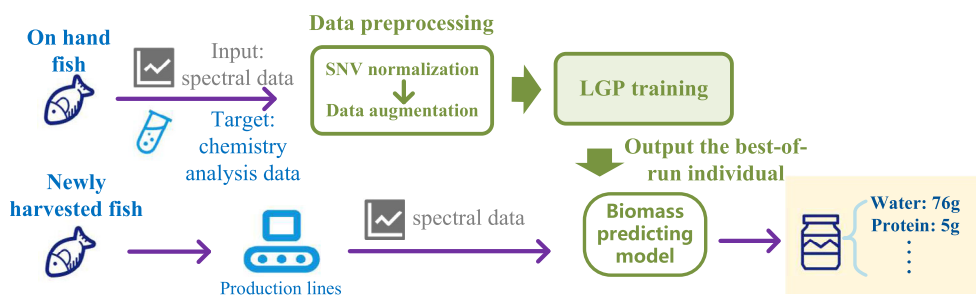


FIGURE 1 | Applying linear genetic programming with tunable primitives (LGP-TP) for fish biomass component prediction. Particularly, LGP-TP synthesizes regression models based on the spectral data (inputs) and the chemical ground truth of on-hand fish samples (target outputs). The synthesized regression model from LGP-TP predicts the components of unseen fish biomass samples in real-world production.

prediction performance over nearly half of the targets and has a better overall performance than the other methods. In addition, the synthesized regression models by LGP-TP have good interpretability. The synthesized models highlight the linear correlation between inputs and targets and specific peaks over the spectral data. Furthermore, our further investigation of other data treatments of fish spectral data and of a popular symbolic regression benchmark, SRBench, shows its potential generality and high training efficiency.

2 | Background

2.1 | Spectral Data Analysis

The main idea of spectral data analysis is to shine a light on matter and detect its physical and chemical characteristics based on the light reflection (Raman and Krishnan 1928). When the laser (photon) hits the molecule of the matter, the molecule exchanges energy with the photon and scatters photons (i.e., reflects light) with certain wavelengths, including the photons with the same energy (i.e., Rayleigh scattering), lower energy (i.e., Stokes Raman scattering), and higher energy (i.e., Anti-Stokes Raman scattering). As molecules of different matters have different characteristics in scattering photons, we can identify different matters based on their scattering patterns. Specifically, this paper focuses on InGaAs Raman spectroscopy data, which is a commonly used spectral wavelength region of scattering photons.

Spectral data analysis has a wide range of applications. As early as 1970s, spectral data analysis has been widely used in collecting environmental and geographic information (Platt and Denman, 1975; Yiou et al. 1996). In recent years, spectral data analysis has been extensively applied to tobacco, oil, and food production (Liu et al. 2024; Qiu et al. 2024; Yang and Hu 2025). However, despite the great success of spectral data analysis, the effectiveness of the analysis is often limited by a small number of data instances and noise among features (Yang and Hu 2025).

2.2 | Symbolic Regression

This paper treats spectral data analysis as a symbolic regression problem. The main goal of symbolic regression is to automatically synthesize a mathematical formula (i.e., symbolic) that can approximate the given data (i.e., regression) (Makke and Chawla 2024; Dong and Zhong 2025). Different from canonical regression that optimizes the parameters of a given model, symbolic regression searches in a symbolic space, consisting of functions (e.g., + and $\times$) and terminals (e.g., x_1 , and π) that finally construct a mathematical formula. This allows symbolic regression to get rid of the assumption on a predefined model and produce mathematic models that better describe the given data.

We analyze the fish spectral data by synthesizing a mathematical model to describe the relationship between spectral data and fish biomass. Because the spectral data usually has a limited number of data instances and a high number of input features, it is non-trivial for human experts to predefine an effective mathematical model, which is suitable for symbolic regression methods to be applied.

2.3 | Linear Genetic Programming

Linear genetic programming (LGP) is under the umbrella of genetic programming methods. The most significant feature of LGP is the linear representation, i.e., representing computer programs by lines of instructions. This leads to advantages over other genetic programming methods, including easy use of control flow operators, multiple outputs, and straightforward theoretical modeling (Brameier and Banzhaf 2007; Huang et al. 2024, 2025). Particularly, in this work, the multiple outputs of LGP facilitate us to design more sophisticated tunable primitives for the prediction of components in fish biomass, rather than unary functions only, and the theoretical modeling on LGP implies the detailed design of tunable primitives.

Figure 2a shows an example of an LGP individual with four instructions. The LGP program initializes the registers (i.e., R[0] to R[2] in the example) with a certain value, such as 0 in our work. The LGP program executes the instructions sequentially. Each instruction accepts the values of source registers on the right, and stores the calculation results to the destination register on the left. The final output is stored in an output register (e.g., the first register R0 in this work). The example individual represents a mathematical formula of $\frac{(x_1 + 1)^2 x_0}{x_1}$.

3 | Proposed Method

3.1 | Overview

Figure 2b shows the training framework of LGP-TP. First, LGP-TP randomly initializes a population of individuals based on the given tunable and basic primitives. These primitives and the LGP representation define our search space. All the individuals are evaluated to predict the given target outputs. The prediction performance of biomass is the fitness of individuals. LGP produces the next-generation population based on the individuals with competitive fitness by four genetic operators (i.e., micro and macro mutation, crossover, and swap). After generations of evolution, LGP-TP outputs the best-of-run individual (i.e., a mathematical model) for unseen spectral data.

Macro mutation produces new programs by adding a new instruction into the LGP program or removing an instruction from the program. The term “macro” indicates that this mutation will change the total number of instructions in a program. In contrast, the micro mutation does not change the total number of instructions. It produces new programs by varying the primitives within an LGP instruction. Both swapping and crossover produce new programs by moving instructions. Specifically, the swapping operator swaps two consecutive instructions within an LGP program, while crossover exchanges two instruction sub-sequences from two different LGP programs (Banzhaf et al. 1998).

Particularly, we tune the tunable primitives in the micro mutation by minimizing the loss function $\mathcal{L}(\mathbf{W})$. As the multivariate linear regression function is always appended at the end of a program, it can be tuned after executing the whole new program. By iteratively varying and keeping better individuals in the LGP population, it is likely for LGP to search for competitive regression models.

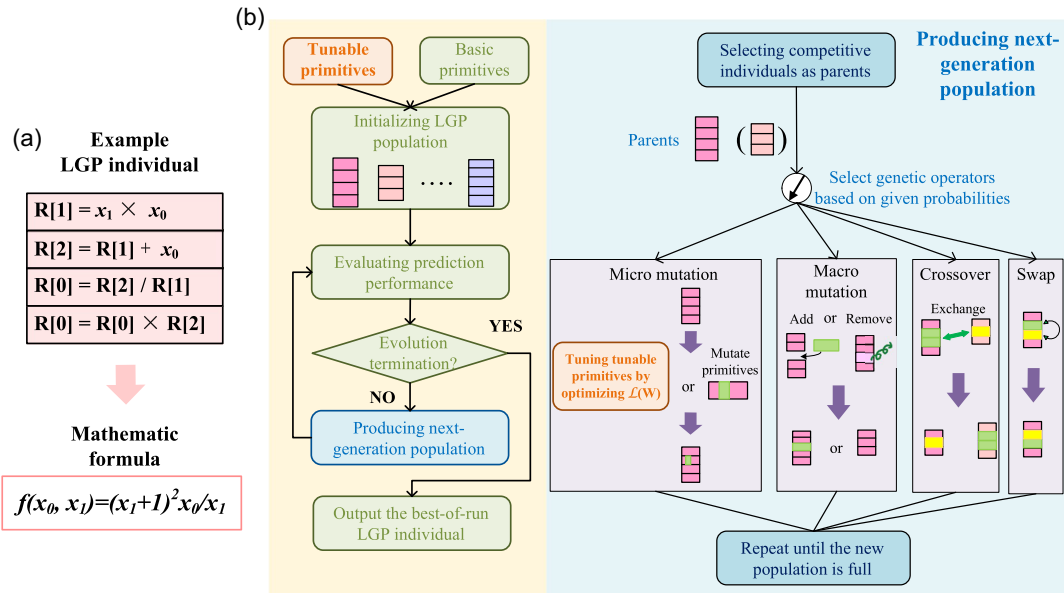


FIGURE 2 | Overview of linear genetic programming with tunable primitives (LGP-TP). (a) An example LGP program, representing a formula of $\frac{(x_1 + 1)^2 x_0}{x_1}$. (b) Training process of LGP-TP: evolutionary framework of LGP-TP (yellow-shaded area), production of the next-generation population (blue-shaded area). The main differences from the basic LGP evolutionary framework are highlighted in orange (i.e., tunable primitives and tuning their parameters).

There are two major differences between LGP-TP and basic LGP, the primitive set and the genetic operator. Specifically, LGP-TP includes three kinds of tunable primitives in its primitive set and tunes the parameters of these tunable primitives in the genetic operator. The tunable primitives record the immediate inputs for themselves during program execution. The genetic operator produces new programs by optimizing the parameters of the tunable primitives, aiming to approximate the target output based on the immediate inputs.

3.2 | Tunable Primitives

Primitives are the elements to construct possible instructions in LGP. The primitives in LGP include terminals (e.g., problem inputs and registers) and functions (e.g., arithmetic operations). Although some existing studies of GP have included numerical numbers in their primitive set (Rovito et al. 2023; Koza 1992; Topchy and Punch 2001), many of them just randomly generate numerical numbers without further tuning and cannot effectively evolve concise regression models with numerical numbers.

This paper proposes evolving tunable primitives in LGP. The main idea of tuning the parameters in the tunable primitives is to approximate the target outputs based on the immediate inputs of the LGP instruction. The major difference of the proposed tunable primitives from the existing constant generation methods is that we embed the search of constants into the search of expression structures. We first randomly generate the coefficients of the primitives (like ephemeral random creation), then tune the coefficients based on the intermediate inputs and target outputs with a restricted step size when breeding offspring. By this means, LGP-TP effectively evolves numerical coefficients within primitives, not just the coefficients wrapping primitives. The tunable primitives help LGP represent a wider range of

functions and reduce the disruption of crossovers by encapsulating complex functions into a single node.

We denote tunable primitives as $\mathcal{T}(\mathbf{X}, \mathbf{W}, \cdot)$, which is essentially a set of functions accepting the immediate input of the LGP instruction $\mathbf{X} = [x_1, x_2, \dots, x_n]^T$, the tunable coefficients $\mathbf{W}$, and other primitive-specific arguments. $\mathbf{X}$ and $\mathbf{W}$ have different dimensions for different primitives. The output of $\mathcal{T}(\mathbf{X}, \mathbf{W}, \cdot)$ is a vector $[\tau(x_1, \mathbf{W}, \cdot), \tau(x_2, \mathbf{W}, \cdot), \dots, \tau(x_n, \mathbf{W}, \cdot)]^T \in \mathbb{R}^n$ where $\tau(x_j, \mathbf{W}, \cdot) j \in [1..n]$ is the element-wise result of $\mathcal{T}$, and n is the number of training instances. We tune $\mathbf{W}$ to minimize the loss function

$$\mathcal{L}(\mathbf{W}) = \|\mathcal{T}(\mathbf{X}, \mathbf{W}, \cdot) - \mathbf{Y}\|^2 = \sum_{j=1}^n (\tau(x_j, \mathbf{W}, \cdot) - y_j)^2, \quad (1)$$

where $\mathbf{Y} = (y_1, y_2, \dots, y_n)^T$ is the target output. There are three types of tunable primitives in our primitive set: tunable terminals, tunable functions, and a multivariate linear regression function. These three types of tunable primitives approximate the target output at the beginning, in the middle, and at the end of a program. To avoid biasing the LGP toward local optima based on incorrect expression structures, we restricted the learning rate of tunable primitives, while simultaneously applying the mutation and crossover to search the expression structures.

3.2.1 | Tunable Terminals

We propose ten tunable primitives, including linear-regression-based terminal (LR), linear regression terminal on 1st-order derivative (1stDLR), average terminal (Avg), standard deviation terminal (Std), fluctuation terminal (Fluctuate), negative-slope terminal (NegSlope), positive-slope terminal (PosSlope), peak terminal (Peak), valley terminal (Valley), and peak location terminal (PeakLoc). All these ten tunable primitives accept

four arguments: the immediate input $\mathbf{X}$, coefficient vector $\mathbf{W}$, and two feature indices α and β . The immediate input of tunable terminals are features of data instances. α and β specify a range of consecutive input features $\mathbf{X}_{\alpha:\beta}$ to make use of the correlation among adjacent features. α and β are searched as ephemeral random constants in GP (Koza 1992). Specifically, we define the element-wise form $\tau(\mathbf{x}_j, \mathbf{W}, \cdot)$ of the ten tunable terminals as follows.

LR approximates the target outputs based on a range of input features. $\text{LR}(\mathbf{x}_j, \mathbf{W}_L, \alpha, \beta) : \mathbb{R}^{\beta-\alpha+1} \times \mathbb{R}^{\beta-\alpha+2} \times \mathbb{R} \times \mathbb{R} \mapsto \mathbb{R}$ indicates that LR performs linear regression over the α^{th} to β^{th} features $(\mathbf{x}_j)_{\alpha:\beta}$ with a vector of coefficients $\mathbf{W}_L$. Specifically,

$$\text{LR}(\mathbf{x}_j, \mathbf{W}_L, \alpha, \beta) = [1, (\mathbf{x}_j)_{\alpha:\beta}^T] \cdot \mathbf{W}_L,$$

where $[1, (\mathbf{x}_j)_{\alpha:\beta}^T]$ is a vector with an augmented constant of 1. We tune $\mathbf{W}_L$ to optimize Equation (1) via the linear least square regression method, as shown in Equation (2).

$$\hat{\mathbf{W}}_L = (\mathbf{X}_L^T \mathbf{X}_L)^{-1} \mathbf{X}_L^T \mathbf{Y}, \quad (2)$$

where the first column of $\mathbf{X}_L : = [1, \mathbf{X}_{\alpha:\beta}]$ is all ones.

$\text{1stDLR}(\mathbf{x}_j, \mathbf{W}_D, \alpha, \beta) : \mathbb{R}^{\beta-\alpha+1} \times \mathbb{R}^{\beta-\alpha+1} \times \mathbb{R} \times \mathbb{R} \mapsto \mathbb{R}$ approximates the target outputs by linear regression based on the first-order derivative of $(\mathbf{x}_j)_{\alpha:\beta}$.

$$\text{1stDLR}(\mathbf{x}_j, \mathbf{W}_D, \alpha, \beta) = [1, [(\mathbf{x}_j)_{\alpha+1:\beta} - (\mathbf{x}_j)_{\alpha:\beta-1}]^T] \cdot \mathbf{W}_D,$$

where $[(\mathbf{x}_j)_{\alpha+1:\beta} - (\mathbf{x}_j)_{\alpha:\beta-1}]^T$ indicates the first-order derivative of $(\mathbf{x}_j)_{\alpha:\beta}$. We tune $\mathbf{W}_D$ to approximate the target output $\mathbf{Y}$ via the linear least square regression method, as shown in Equation (3) where $\mathbf{X}_D : = [1, \mathbf{X}_{\alpha+1:\beta} - \mathbf{X}_{\alpha:\beta-1}]$.

$$\hat{\mathbf{W}}_D = (\mathbf{X}_D^T \mathbf{X}_D)^{-1} \mathbf{X}_D^T \mathbf{Y}. \quad (3)$$

The rest of the six tunable terminals follow the same form of

$$\tau(\mathbf{x}_j, \mathbf{W}, \alpha, \beta) = [1, \gamma((\mathbf{x}_j)_{\alpha:\beta})] \mathbf{W}_a,$$

where $\gamma((\mathbf{x}_j)_{\alpha:\beta}) : \mathbb{R}^{\beta-\alpha+1} \mapsto \mathbb{R}$ represents a certain kind of information over the α^{th} to β^{th} input features $(\mathbf{x}_j)_{\alpha:\beta}$. Specifically, $\gamma((\mathbf{x}_j)_{\alpha:\beta})$ can be:

1. $\text{Avg}((\mathbf{x}_j)_{\alpha:\beta}) = \frac{1}{\beta-\alpha+1} \sum_{i=\alpha}^{\beta} (\mathbf{x}_j)_i$ returns the average value of the α^{th} to β^{th} input features. $(\mathbf{x}_j)_i$ is the i^{th} feature of input $\mathbf{x}_j$.
2. $\text{Std}((\mathbf{x}_j)_{\alpha:\beta}) = \sqrt{\frac{1}{\beta-\alpha+1} \sum_{i=\alpha}^{\beta} ((\mathbf{x}_j)_i - \text{Avg}((\mathbf{x}_j)_{\alpha:\beta}))^2}$ returns the standard deviation of the α^{th} to β^{th} input features.
3. $\text{Fluctuate}((\mathbf{x}_j)_{\alpha:\beta}) = \frac{1}{\beta-\alpha} \sum_{i=\alpha+1}^{\beta} |(\mathbf{x}_j)_i - (\mathbf{x}_j)_{i-1}|$ returns the average accumulated variation over the α^{th} to β^{th} input features.
4. $\text{NegSlope}((\mathbf{x}_j)_{\alpha:\beta}) = \frac{1}{\beta-\alpha} \sum_{i=\alpha+1}^{\beta} \min\{(\mathbf{x}_j)_i - (\mathbf{x}_j)_{i-1}, 0\}$ and $\text{PosSlope}((\mathbf{x}_j)_{\alpha:\beta}) = \frac{1}{\beta-\alpha} \sum_{i=\alpha+1}^{\beta} \max\{(\mathbf{x}_j)_i - (\mathbf{x}_j)_{i-1}, 0\}$

return the average accumulated decrement and increment, respectively.

5. $\text{Peak}((\mathbf{x}_j)_{\alpha:\beta}) = \max_{i \in [\alpha, \beta]} (\mathbf{x}_j)_i$ and $\text{Valley}((\mathbf{x}_j)_{\alpha:\beta}) = \min_{i \in [\alpha, \beta]} (\mathbf{x}_j)_i$ return the maximum and minimum value over the α^{th} to β^{th} input features, respectively.
6. $\text{PeakLoc}((\mathbf{x}_j)_{\alpha:\beta}) = \arg\max_{i \in [\alpha, \beta]} (\mathbf{x}_j)_i$ returns the feature index of the maximum value over α to β .

Let $\{\gamma((\mathbf{x}_j)_{\alpha:\beta})\}_n$ be an n -dimension vector, we have $\Gamma : = [1, \{\gamma((\mathbf{x}_j)_{\alpha:\beta})\}_n] \in \mathbb{R}^{n \times 2}$ whose first column is all ones. We optimize Equation (1) by tuning $\mathbf{W}_a$ via the linear least square regression method.

$$\hat{\mathbf{W}}_a = (\Gamma^T \Gamma)^{-1} \Gamma^T \mathbf{Y}.$$

To reduce the search space, the range $[\alpha, \beta]$ covers up to 50% of the total features for fish biomass prediction and up to 10% of features for the SRBench problems.

3.2.2 | Tunable Functions

We propose four tunable functions, including the linear-regression-based function (LRF), tunable trigonometric function (SinRF), tunable exponential function (ExpORF), and tunable polynomial function (PowRF). These functions (e.g., “sin” and “exp”) are all commonly used functions in existing symbolic regression studies. All the four tunable functions are unary functions for the sake of simplicity, accepting one input value x (the intermediate results of the LGP program). The immediate inputs for tunable functions are recorded during program execution. To ensure the outputs of tunable functions better approximate the target outputs, the tunable functions approximate the residual error between the immediate input x_j and the target output y by adding the raw input x_j . We define the element-wise form $\tau(\mathbf{x}_j, \mathbf{W}, \cdot)$ of the four tunable functions as follows.

1. $\text{LRF}(x_j, [\omega_0, \omega_1]^T) = \omega_0 + \omega_1 x_j$.
2. $\text{SinRF}(x_j, [\omega_0, \omega_1, \omega_2, \omega_3]^T) = \omega_0 + \omega_1 \sin(\omega_2 x_j + \omega_3) + x_j$.
3. $\text{ExpORF}(x_j, [\omega_0, \omega_1]^T) = \omega_0 + (\omega_1^2 + 1)^{x_j} + x_j$.
4. $\text{PowRF}(x_j, [\omega_0, \omega_1]^T) = \omega_0 + |x_j|^{\omega_1} + x_j$.

Let $\{\text{LRF}(x_j, [\omega_0, \omega_1]^T)\}_n$ be an n -dimension vector and $\mathbf{X}_F : = [1, \{\text{LRF}(x_j, [\omega_0, \omega_1]^T)\}_n] \in \mathbb{R}^{n \times 2}$ be a matrix whose first column is all ones, we tune the parameters of LRF using the least square regression method.

$$[\hat{\omega}_0, \hat{\omega}_1]^T = (\mathbf{X}_F^T \mathbf{X}_F)^{-1} \mathbf{X}_F^T \mathbf{Y}.$$

We tune the parameters of the other three tunable functions (SinRF, ExpORF, PowRF) by a gradient descent method with a step size λ ($\lambda = 0.1$ in our work) to optimize Equation (1). Specifically, we have

$$\hat{\omega} = \omega - \lambda \frac{\partial \mathcal{L}}{\partial \omega} / \left\| \frac{\partial \mathcal{L}}{\partial \omega} \right\|,$$

and have the following partial derivatives for sinRF:

$$\frac{\partial \mathcal{L}}{\partial \omega_0} = \sum_j^n 2(\omega_0 + \omega_1 \sin(\omega_2 x_j + \omega_3) + x_j - y_j),$$

$$\frac{\partial \mathcal{L}}{\partial \omega_1} = \sum_j^n 2(\omega_0 + \omega_1 \sin(\omega_2 x_j + \omega_3) + x_j - y_j) \omega_1 \sin(\omega_2 x_j + \omega_3),$$

$$\frac{\partial \mathcal{L}}{\partial \omega_2} = \sum_j^n 2(\omega_0 + \omega_1 \sin(\omega_2 x_j + \omega_3) + x_j - y_j) \omega_1 \cos(\omega_2 x_j + \omega_3) x_j,$$

$$\frac{\partial \mathcal{L}}{\partial \omega_3} = \sum_j^n 2(\omega_0 + \omega_1 \sin(\omega_2 x_j + \omega_3) + x_j - y_j) \omega_1 \cos(\omega_2 x_j + \omega_3).$$

For ExpORF:

$$\frac{\partial \mathcal{L}}{\partial \omega_0} = \sum_j^n 2(\omega_0 + (\omega_1^2 + 1)^{x_j} + x_j - y_j),$$

$$\frac{\partial \mathcal{L}}{\partial \omega_1} = \sum_j^n 2(\omega_0 + (\omega_1^2 + 1)^{x_j} + x_j - y_j) 2\omega_1 x_j (\omega_1^2 + 1)^{x_j - 1}.$$

For PowRF:

$$\frac{\partial \mathcal{L}}{\partial \omega_0} = \sum_j^n 2(\omega_0 + |x_j|^{\omega_1} + x_j - y_j),$$

$$\frac{\partial \mathcal{L}}{\partial \omega_1} = \sum_j^n 2(\omega_0 + |x_j|^{\omega_1} + x_j - y_j) |x_j|^{\omega_1} \ln(|x_j|).$$

The parameters ω in the tunable functions have a limited value range of $[-3, 3]$.

3.2.3 | Multivariate Linear Regression Function

The proposed multivariate linear regression function (MVLRL) is a special tunable function that is always appended at the end of LGP programs. It integrates the values from multiple registers into one output value (i.e., storing the result in an output register). To keep LGP programs concise, each LGP program only has one MVLRL function for each output register. We define the element-wise form of MVLRL as follows.

$$\text{MVLRL}(\mathbf{x}_j, \mathbf{W}_M) : \mathbb{R}^r \times \mathbb{R}^{r+1} \mapsto \mathbb{R} = [\mathbf{1}, \mathbf{x}_j^T] \cdot \mathbf{W}_M,$$

where r is the number of registers inputted to MVLRL. $\mathbf{W}_M$ is tuned based on the register values after executing the precedent instructions in the LGP program. MVLRL accepts $r = 20$ register values for fish spectroscopic data ($\mathbf{x}_j \in \mathbb{R}^{20}$) and accepts $r = 4$ register values for SRBench problems ($\mathbf{x}_j \in \mathbb{R}^4$). We tune $\mathbf{W}_M$ by optimizing Equation (1) via the least square regression method where $\mathbf{X}_M := [\mathbf{1}, \mathbf{X}] \in \mathbb{R}^{n \times (1+r)}$ with the first column of all ones.

$$\hat{\mathbf{W}}_M = (\mathbf{X}_M^T \mathbf{X}_M)^{-1} \mathbf{X}_M^T \mathbf{Y}.$$

3.2.4 | Overall Primitive Set

The overall primitive set includes the proposed tunable primitives and basic arithmetic operations ($+$, $-$, $\times$, Analytic Quotient), trigonometric operations ($\sin$, $\cos$, $\tanh$), boolean operations ($\max$, $\min$), and others ($\sqrt{x}$, x^2 , $\exp(x)$, and $\ln(|x|)$). These primitives

consist of all possible instructions together with registers. In this work, an LGP instruction includes one destination register, one function, and two terminals such as problem inputs and source registers.

4 | Experiment Design

4.1 | Fish Spectroscopic Dataset

We built a fish spectroscopic dataset for this study, covering two deep-water fish species in New Zealand, Hoki and Mackerel. First, to avoid enzymatic and oxidative degradation, fish samples are supplied fresh, carefully subsampled, and frozen once they are harvested. To enable cross-domain research in the future, we label fish samples with as much metadata as possible, including their source locations and the sampling time of year. After that, we measure their chemical characteristics and spectroscopic features, respectively. To reduce measuring bias, each fish sample was measured three times, resulting in a total of 117 spectroscopic instances. Since the data collection process is cumbersome and expensive, the number of instances is small from a machine learning perspective. However, this dataset represents the most extensively characterized fish chemistry (to our knowledge) in existence.

To verify the performance of the proposed method, we apply LGP-TP to predict 10 kinds of biomass based on the spectroscopic data. The biomass targets included water, ash, nitrogen, protein, fat, lipids yield, wax/sterol esters, triacylglycerols, free fatty acids, and sterols. We applied InGaAs Raman at 1064 nm to obtain the spectroscopic data and truncated the wavenumber range of 580.109 to 202.533 (cm^{-1}) to reduce the impact of glass vials of fish meat. As a result, the obtained spectroscopic data had a wavenumber range of 1891.58–580.109 (cm^{-1}), with 427 input features. We corrected the obtained spectroscopic data by standard normal variate transformation (SNV).

Considering the limited spectroscopic instances in the dataset, we applied sixfold cross-validation to evaluate machine learning models. Specifically, we split the 117 instances into sixfolds, each fold with approximately 20 instances (6 or 7 fish samples). In one sixfold cross-validation, a machine learning model alternatively took one fold of split instances as unseen test data and trained on the other five folds, resulting in six different training (and test) performances. The mean of the six different training (and test) performances is regarded as the performance of one sixfold cross-validation. Each machine learning model ran the sixfold cross-validation for 10 times with unique random seeds. To enhance the performance of machine learning models, we augmented the training data of machine learning models 50 times (i.e., a total of approximately 5000 training instances) by three strategies: spectral data augmentation (Bjerrum et al. 2017), mix-up (i.e., linear interpolation) (Zhang et al. 2018), and Gaussian noise. These three strategies produced 50%, 25%, and 25% of new instances, respectively.

4.2 | Comparison Design

We compared LGP-TP with six popular machine learning methods for predicting fish biomass, including partial least squares regression

(PLSR) (Wold et al. 2001), K-nearest neighbor (KNN), multi-layer perceptron (MLP), gradient boosting regression (XGB), random forest regression (RF), and FishCNN (Zhou et al. 2024). Particularly, PLSR is one of the commonly used machine learning methods for analyzing spectroscopic data, and FishCNN is one of the state-of-the-art methods tailored toward fish biomass prediction based on Raman spectral data. All seven methods ran 10 times sixfold cross-validation for each biomass target, respectively.

The parameters of all the compared methods are summarized in Table 1. The parameters of PLSR and FishCNN follow their recommended settings in (Zhou et al. 2024), and the parameters of the other methods are the default settings. Specifically, PLSR has 10 components, KNN has $k = 3$, MLP, whose activation units are “Relu”, learns with a solver of “adam” and with a learning rate of $1e^{-4}$ for 1000 iterations, and XGB and RF have up to 100 decision trees (“n_estimators”), each with a maximum depth of 5. FishCNN is trained by weighted adam (“AdamW”) for up to 1500 epochs. FishCNN has two convolutional layers, each with the same architecture, and three fully connected layers. The parameters of LGP-TP mostly follow the existing studies (Huang et al. 2024). LGP-TP evolves a population of 250 individuals for 100 generations. Each LGP-TP individual has up to 50 instructions with 30 available registers. LGP-TP applies the macro mutation, the linear crossover operator, and the reproduction operator with operator rates of 0.6, 0.3, and 0.1, respectively. The swap operator and micro mutation are applied after each macro mutation and linear crossover. In each offspring breeding, mutation operators (i.e., macro and micro mutation) can change up to one instruction, and the linear crossover can swap subprograms with up to 30 instructions.

5 | Experiment Results

5.1 | Prediction Performance

The main aim of this paper is to automatically construct effective regression models by LGP-TP for predicting fish biomass for newly harvested fish. We evaluated the prediction performance based on the R^2 values over test instances. Figure 3 shows the distributions of test R^2 over the 10 sixfold cross-validations of the compared methods. We can see that the proposed LGP-TP had the highest test R^2 distributions in the biomass targets of water, ash, and lipids

yield, and has a very competitive performance with other compared methods in nitrogen, fat, and triacylglycerol. To statistically analyze the performance of LGP-TP, we applied Friedman’s test with a significance level of 0.05 and with the Bonferroni correction and the Wilcoxon rank-sum test with a significance level of 0.05 to the test R^2 over the 10 sixfold cross-validation. The Friedman test had a p -value of 0.0061, indicating a significant difference among the compared methods. The radar chart in Figure 3 shows the mean ranks of the compared methods by Friedman’s test over the ten biomass targets. LGP-TP (the red polygon) covered the largest area on the radar chart. Specifically, LGP-TP had the best mean ranks in the biomass targets such as water, lipid yield, and free fatty acids. The legend in Figure 3 summarizes the average of the mean ranks and the results of the Wilcoxon rank-sum test. “win” indicates the number of targets where a compared method is significantly better than the proposed LGP-TP. “draw” indicates the number of targets without significantly different test R^2 . “lose” indicates the number of targets where a compared method is significantly worse than LGP-TP. We can see that the other six compared methods performed significantly worse than LGP-TP on at least four biomass targets. Specifically, the second-best and third-best compared methods (i.e., FishCNN and XGB) were significantly worse than LGP-TP on four targets with p -value < 0.0162 and p -value $< 3e^{-4}$, respectively, indicating significant superiority of LGP-TP over its opponents. In addition, the mean rank of LGP-TP (5.45) had a large gap compared with the second- and third-highest mean ranks (i.e., FishCNN (4.65) and XGB (4.6)). These results highlight the good overall performance of LGP-TP for predicting different kinds of biomass based on fish spectroscopic data.

5.2 | Constraints on Predicting Fish Biomass via Spectral Data

Figure 3 shows negative R^2 values in low-content biomass targets, i.e., triacylglycerol, free fatty acids, and sterols. The prediction performance on these targets is quite sensitive to the noise in chemical analyses. Almost all machine learning methods, including the standard method (i.e., PLSR) and the state of the art (i.e., FishCNN), fail to get positive R^2 results on the three targets due to the low concentration of these targets. The negative R^2 results imply that when compounds (e.g., free fatty acids and sterols) are too low in concentration, machine learning models

TABLE 1 | Key parameters of comparison methods.

Methods	Key parameters
Partial Lease Squares Regression	n_components = 10
K-nearest Neighbor	K = 3
Multi-layer Preceptron	max_iter = 1000, solver = ‘adam’, activation=‘Relu’, alpha = 0.0001
Gradient Boosting Regression	max_depth = 5, n_estimators = 100
Random Forest	max_depth = 5, n_estimators = 100
FishCNN	max epochs = 1500, optimizer = ‘AdamW’, 2 × Convolutional layers (16 filters, 64 filter size, 1 stride), fully connected layers = [128,16,3]
Linear Genetic Programming with Tunable Primitives	popsiz = 250, n_gen = 100, max_programsize = 50, n_registers = 30, macro_mut: crossover:reproduction rate = 0.6:0.3:0.1

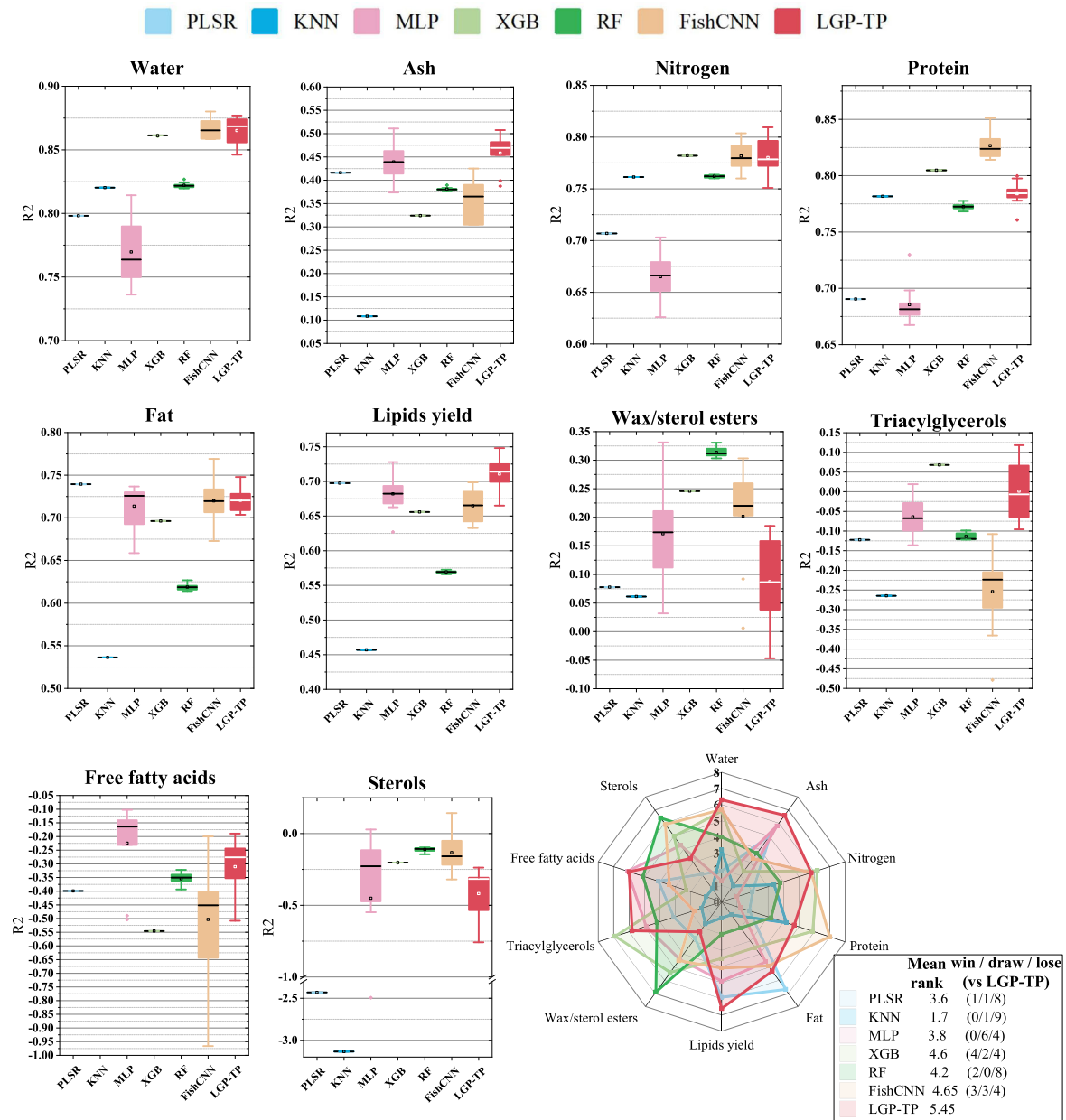


FIGURE 3 | Test R^2 of the compared methods in the 10 sixfold cross-validations, each box plot for a biomass target. The radar chart shows the mean ranks of the comparison methods over different biomass targets. The legend of the chart shows the overall mean ranks by Friedman's test and the statistical results by the Wilcoxon rank-sum test ($\alpha=0.05$).

cannot reliably predict the fish biomass composition through spectroscopy data. However, we can find that the proposed LGP-TP is yet more resistant to the noise than many other compared methods in these low-content targets. For example, LGP-TP takes a leading position on triacylglycerol and free fatty acids.

To give a clear guide on the condition of reliable predictions, we investigate the impact of the concentration on R^2 values. Figure 4 shows the test R^2 values of all the comparison methods over the average biomass weight of different biomass components. We can see that the test R^2 consistently goes down with the average biomass weight. Particularly, when the biomass weight is lower than 2.5% (i.e., low-concentrated components), there is a significant drop in the test R^2 values for all the machine

learning models. This result suggests that spectral data is more reliable for predicting fish biomass components with a concentration larger than 2.5% biomass weight.

5.3 | Ablation Study

Figure 5 shows the results of our ablation study. To verify the effectiveness of the three kinds of tunable primitives introduced into basic LGP, we compared LGP-TP with basic LGP (i.e., without tunable primitives) and two crippled variants of LGP-TP. The two crippled variants used tunable terminals only or used both tunable terminals and functions, but without the multivariate linear regression function. The two crippled variants are denoted as LGP-TT and LGP-TTF, respectively. The area on the radar

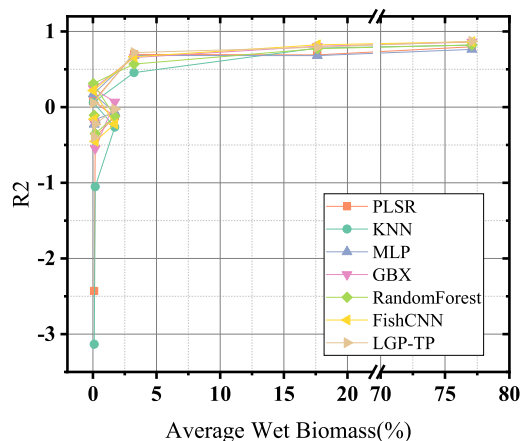


FIGURE 4 | Test R^2 values over biomass weights of the comparison methods.

chart from basic LGP to LGP-TP grew consistently. The legend in Figure 5 shows the statistical results of Friedman's and Wilcoxon rank-sum tests, whose settings are the same as in Figure 3. The overall p -value of Friedman's test of the ablation study was 0.0062, indicating a significant difference in the ablation study. Figure 5 shows growing mean ranks from LGP-TT to LGP-TP. Particularly, LGP-TP significantly improved the overall performance of basic LGP with a p -value = 0.019. The tunable primitives helped LGP perform significantly better in more biomass targets based on the Wilcoxon rank-sum test, i.e., LGP-TT, LGP-TTF, and LGP-TP performed significantly better than basic LGP on 4, 5, and 6 biomass targets, respectively, with p -value < 0.001. Furthermore, LGP-TT and LGP-TTF, which go without the proposed multivariate linear regression function, still outperform the basic LGP in most cases. This implies that the proposed tunable terminals and functions potentially help LGP address the constant generation issue. The ablation study confirms that the proposed tunable primitives enhance the LGP performance for fish biomass prediction.

A noteworthy result is that the basic LGP outperforms all the LGP methods with tunable primitives in three fish components,

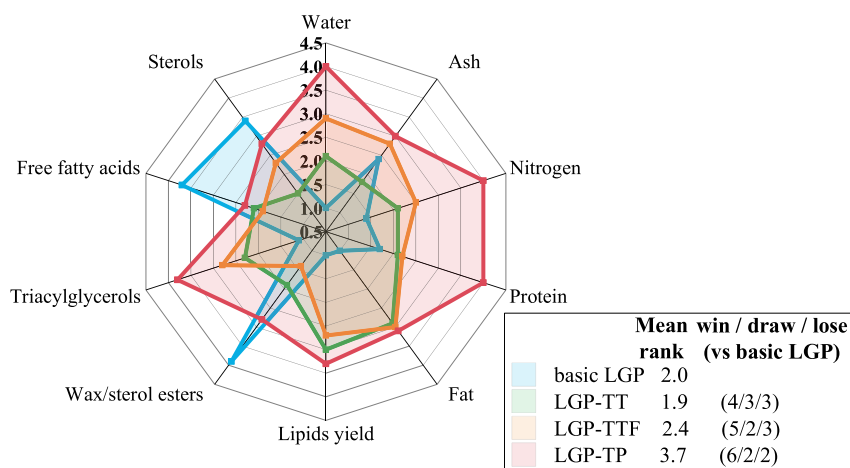


FIGURE 5 | The ablation study of LGP-TP by adding tunable primitives one-by-one in LGP evolution. The legend follows the same design as Figure 3.

i.e., sterols, free fatty acids, and wax/sterol esters. The performance differences between the basic LGP and LGP-TP stem from the data noise. These fish components occur at a low concentration in fish meat (average wet biomass $\leq 2.5\%$ as mentioned in Section 5.2), which includes noise in target outputs. Therefore, the basic LGP, whose learning ability is more restricted by the fixed primitives, is less affected by the noise in training data.

5.4 | Interpretability Analysis

LGP-TP assists humans in understanding spectral data by synthesizing symbolic regression models. The symbolic regression model allows us to infer and reason about the behaviors of the regression model. Figure 6a shows the directed acyclic graph of an example regression model for predicting water in fish meat. This regression model is the shortest program among all the independent runs to facilitate analysis. For an amplified version of Figure 6, please refer to the supplement. The tunable terminals and functions are highlighted in pink and yellow, respectively. The graph nodes also show the input ranges of tunable terminals and the parameters of tunable functions within the square brackets. The multivariate linear regression function is split into "x", "+", and coefficients in the top of the graph. The regression model contained a larger number of tunable terminals and functions, such as "LR" and "SinRF", implying an important role of tunable primitives in the regression model. Particularly, the example regression model preferred tunable terminals like "LR", "Std", and "Avg". These three kinds of features might have a high correlation with the water measurement. The example regression model preferred tunable functions like "SinRF" and "LRF" as well. Regarding the multivariate linear regression function, the example model contained a large number of negative coefficients of multiplication, implying a negative correlation between the wavenumber features and the water measurement.

The synthesized regression models highlighted some interesting characteristics of fish spectroscopic data. Figure 6b shows the highlighted features and the frequency of tunable terminals. Specifically, the lower part of Figure 6b shows the frequency heat maps of tunable terminals over the 10 sixfold cross-validation for

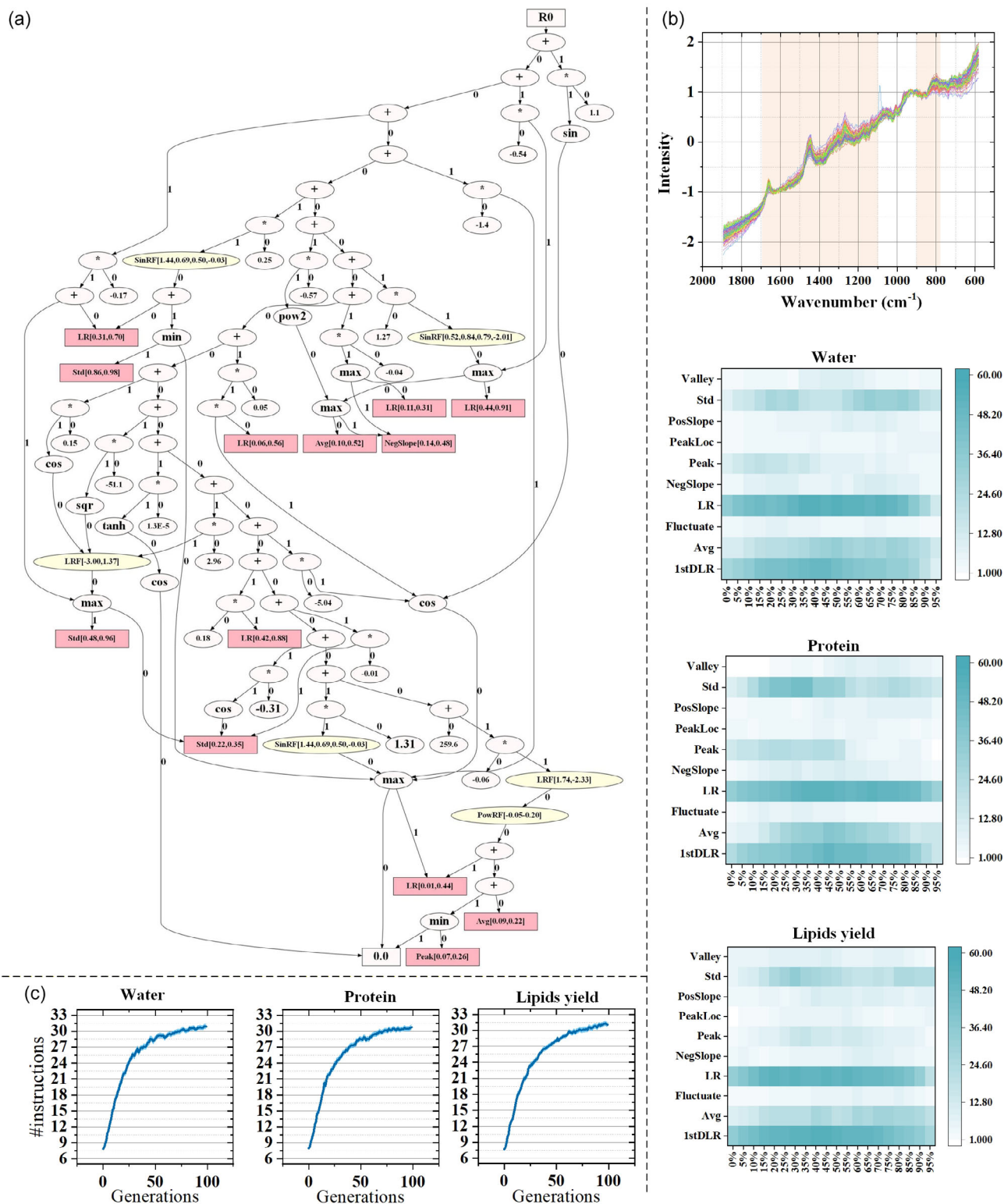


FIGURE 6 | Interpretability analysis of LGP-TP. (a) An example synthesized regression model by LGP-TP for predicting water in fish meat, outputting results from the first register R0. (b) Highlighted input features by the result models. The upper part of sub-figure b is the Raman spectra data with highlighted features in yellow. The three heat maps in the lower part show the frequency of input terminals over feature ranges (X-axis: percentage over the range of wavenumber, Y-axis: tunable terminals). The dark color indicates a high frequency over the 10 sixfold cross-validation. (c) Mean program size ($\pm$ std.) of synthesized regression models over generations from 10 sixfold cross-validation. We denote the number of effective instructions as the program size.

predicting three example biomass targets. The X-axis of the heat maps is the percentage over the wavenumber range. We selected three biomass targets, water, protein, and lipids yield, as examples

to investigate the highlighted features. The three biomass targets are primary biomass in fish meat and usually account for more than 90% of the total weight of a fish. We can see that these three

biomass targets share a similar pattern of frequency heat maps. “LR”, “1stDLR”, “Avg”, “Std”, and “Peak” were the five most frequently used tunable terminals. Particularly, “LR” and “1stDLR” made use of nearly the whole range of wavenumber, while “Std” and “Avg” only preferred roughly 15%–50% and 75%–85%. The upper part of Figure 6 summarizes these overlapped ranges over the spectrum. We can see that the tunable primitives mainly focused on the first three and the last peaks over the spectroscopic data, implying a high correlation of these peaks with fish biomass.

The synthesized models are compact for interpretation. Figure 6c shows the average program size (and standard deviation) of the synthesized models, represented by the number of effective instructions in LGP programs. In the course of LGP-TP evolution, the program size grew from about 5–30 instructions in the three example biomass targets. This confirms that LGP-TP consistently synthesizes effective and compact programs. In addition, the synthesized regression model is less than 30 KB on average, which is acceptable for human analysis.

5.5 | Ability in Avoiding Overfitting

Limited quality and quantity of data easily lead to the overfitting issue in machine learning methods. LGP-TP relieves the overfitting by synthesizing necessarily competitive regression models. To verify the ability of avoiding overfitting in LGP-TP, Figure 7 shows the training and test performance over generations of LGP-TP in four example biomass targets. For the training and test performance of LGP-TP in all the ten biomass targets, refer to the supplement. Specifically, the orange curves indicate the training R^2 of LGP-TP, and the red curves indicate its test R^2 . We can see that the training R^2 of LGP-TP consistently rises over its evolution in all the four biomass targets, and the test R^2 rises or maintains at a certain level. We believe that there is no significant overfitting over LGP-TP evolution since the gap between training and test R^2 does not widen.

5.6 | Generality Across Spectral Data Treatments

The proposed LGP-TP has a good generality across data treatments of spectral data. Good generality of data treatments is

important in practice as production lines might only handle certain data treatments other than the one used in our experiments. We verified the generality of LGP-TP over three more data treatment techniques: 1) InGaAs Raman + LB, 2) FT Raman + SNV, and 3) InGaAs SNV + 1st Der + SW17. Specifically,

1. “InGaAs Raman + LB” applied linear baseline to process the InGaAs Raman at 1064 nm with the wavenumber range of 580.109–202.533 (cm^{-1});
2. “FT Raman + SNV” indicated Fourier transform (FT) Raman processed by SNV and with the wavenumber range of 4001.81–202.533 (cm^{-1});
3. “InGaAs SNV + 1st Der + SW17” indicated the InGaAs Raman at 1064nm with the wavenumber range of 580.109–202.533 (cm^{-1}), processed by SNV, transformed into their 1st-order derivative, and smoothed by a sliding window with size of 17 units.

These data treatments lead to different wavenumber ranges and shapes of spectral data. All these data treatments are split into sixfolds for cross-validation and augmented by the method mentioned in Section 4.1.

We verified the performance of LGP-TP via the comparison with PLSR, XGB, and support vector regression (SVR) (Ahmed et al. 2023). We selected the water, protein, and lipid yield, as example biomass targets to show the test R^2 distribution of the compared methods, as shown in Figure 8. Figure 8 also shows the mean ranks by Friedman’s test of the comparison methods over all the ten biomass targets. LGP-TP had a very competitive test performance compared with the other methods. It achieved the best or second-best mean performance in most cases. The mean ranks by Friedman’s test over the ten biomass targets confirmed this observation. LGP-TP had the best or second-best mean ranks in different data treatments. Particularly, the second-best mean ranks in “FT Raman + SNV” and “InGaAs SNV + 1st Der + SW17” were very close to the best mean rank. On the contrary, the performance of the other methods varied among the three data treatments. For example, XGB ranked third in “InGaAs Raman + LB”, SVR ranked worst in “InGaAs Raman + LB” and “FT Raman + SNV”, and PLSR ranked worst in “InGaAs SNV + 1st Der + SW17”. The results imply the good generality of LGP-TP to other spectral data treatments.

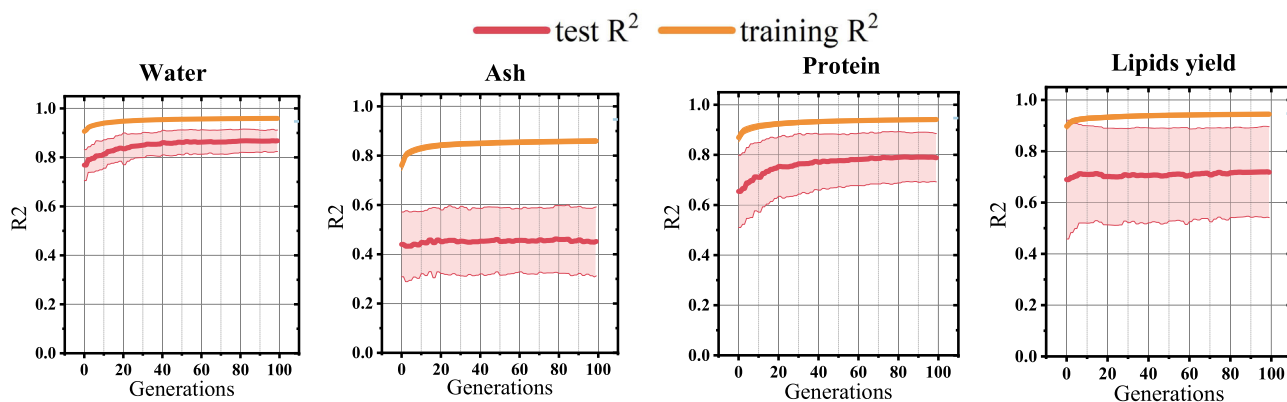


FIGURE 7 | The training and test R^2 of LGP-TP in four example biomass targets.

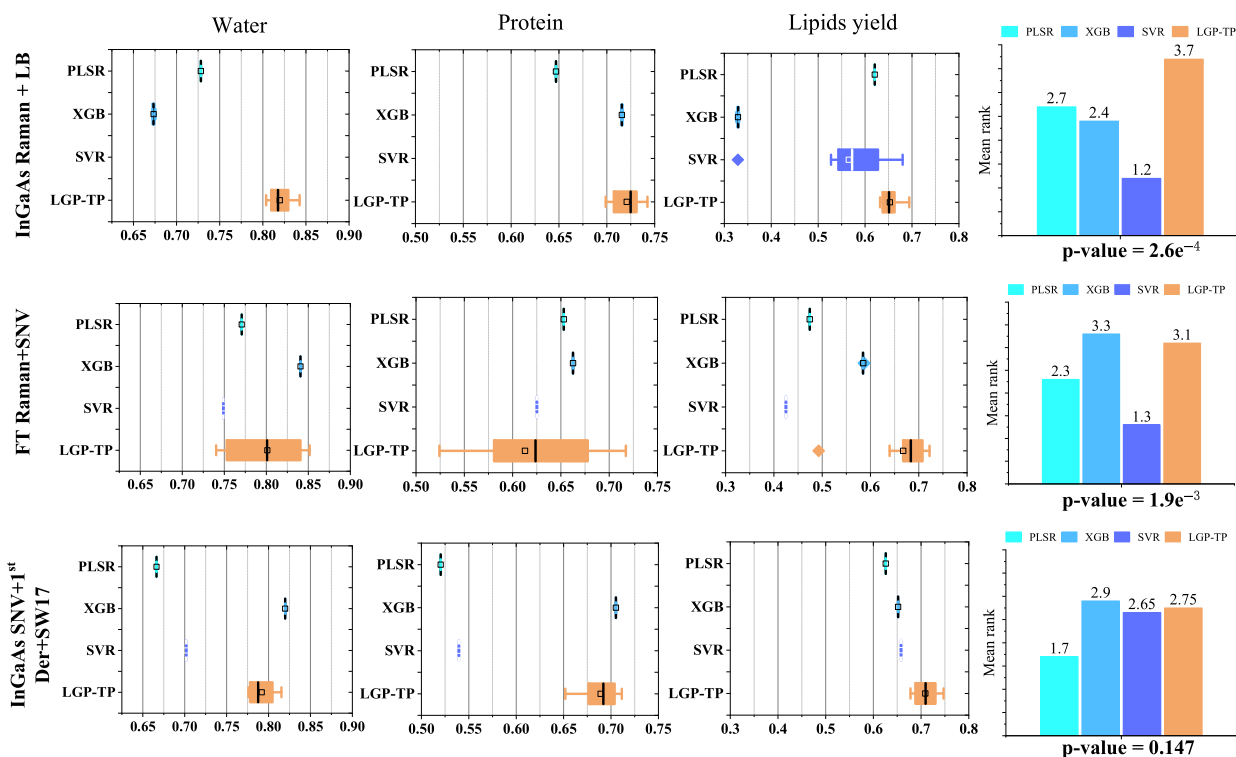


FIGURE 8 | Test R^2 (X-axis) of the four compared methods for different data treatments over 10 sixfold cross-validation. The three rows stand for three different spectral data treatments. We show the box plots of test R^2 on three example biomass targets and the bar chart of mean ranks (and corresponding p -values) over the ten biomass targets by Friedman's test (PLSR: partial least squares regression; XGB: gradient boosting regression; SVR: support vector regression; LGP-TP: linear genetic programming with tunable primitives).

5.7 | Generality Across Regression Problems

The proposed LGP-TP showed a promising effectiveness across regression problems. Although this paper focuses on the fish spectroscopic dataset, we wondered if LGP-TP still has promising performance in other spectroscopic datasets and even a wider range of applications. With this in mind, we extended LGP-TP to a popular symbolic regression benchmark, SRBench (De Franca et al. 2024), to comprehensively investigate the potential generality of the proposed LGP-TP. Specifically, we applied LGP-TP to solve the black-box regression problems in SRBench, a total of 122 problems. These regression problems are mostly collected in practice, covering various application backgrounds and different levels of difficulty. Given that most of the problems in SRBench are with tabular data, we removed the spectroscopic-data-specific tunable terminals (e.g., “1stDLR” and “Std”) and only retained “LR” as tunable terminals. The other tunable functions were kept. The primitive set of LGP-TP for solving SRBench included the raw input features as well. LGP-TP for SRBench evolved 200 generations with a population size of 500 individuals. Each LGP-TP individual had up to 50 instructions and 8 available registers. The multi-variable linear regression function accepted the first four registers as inputs and stored the results in the output register R0. Our experiments were implemented by the Java library ECJ (Luke 2017) and run on AMD EPYC 7702 computing nodes with a single thread, each thread having 16 GB RAM.

We compared LGP-TP with 23 benchmark methods, including PSTree (Zhang et al. 2022), DSR (Petersen et al. 2021), and many other popular regression methods (e.g., XGB and MLP). Figure 9a shows the main results of SRBench. LGP-TP had the second-best test R^2 amongst the benchmark methods and is very competitive with PSTree. LGP-TP likely has a good generality across various domains.

Figure 9a shows that LGP-TP had the smallest model size and training time among the first six methods with the highest test R^2 . Here, we multiplied the number of effective instructions in an LGP program by a factor of 2.0 as the model size, since functions in the primitive set of LGP-TP had up to two inputs. There are three reasons why the introduction of tunable primitives does not lead to increased model size in Figure 9a. First, we fix a small maximum program size (i.e., 50 LGP instructions) in LGP-TP for SRBench. Second, the tunable primitives help LGP-TP to represent more complex functionality with fewer instructions. Third, LGP-TP only produces one model for prediction, rather than a set of models for further ensembling. This implies that LGP-TP synthesizes concise yet very effective regression models with a short training time.

Figure 9b shows the non-dominated sets for optimizing two minimization objectives: model size rank and test R^2 rank. The proposed LGP-TP is in the middle position of the non-dominated set over all the benchmark methods, implying a good balance of LGP-TP between the two conflicting objectives, model size, and model effectiveness. Figure 9c shows the training time increment with the number of training instances. We can see that

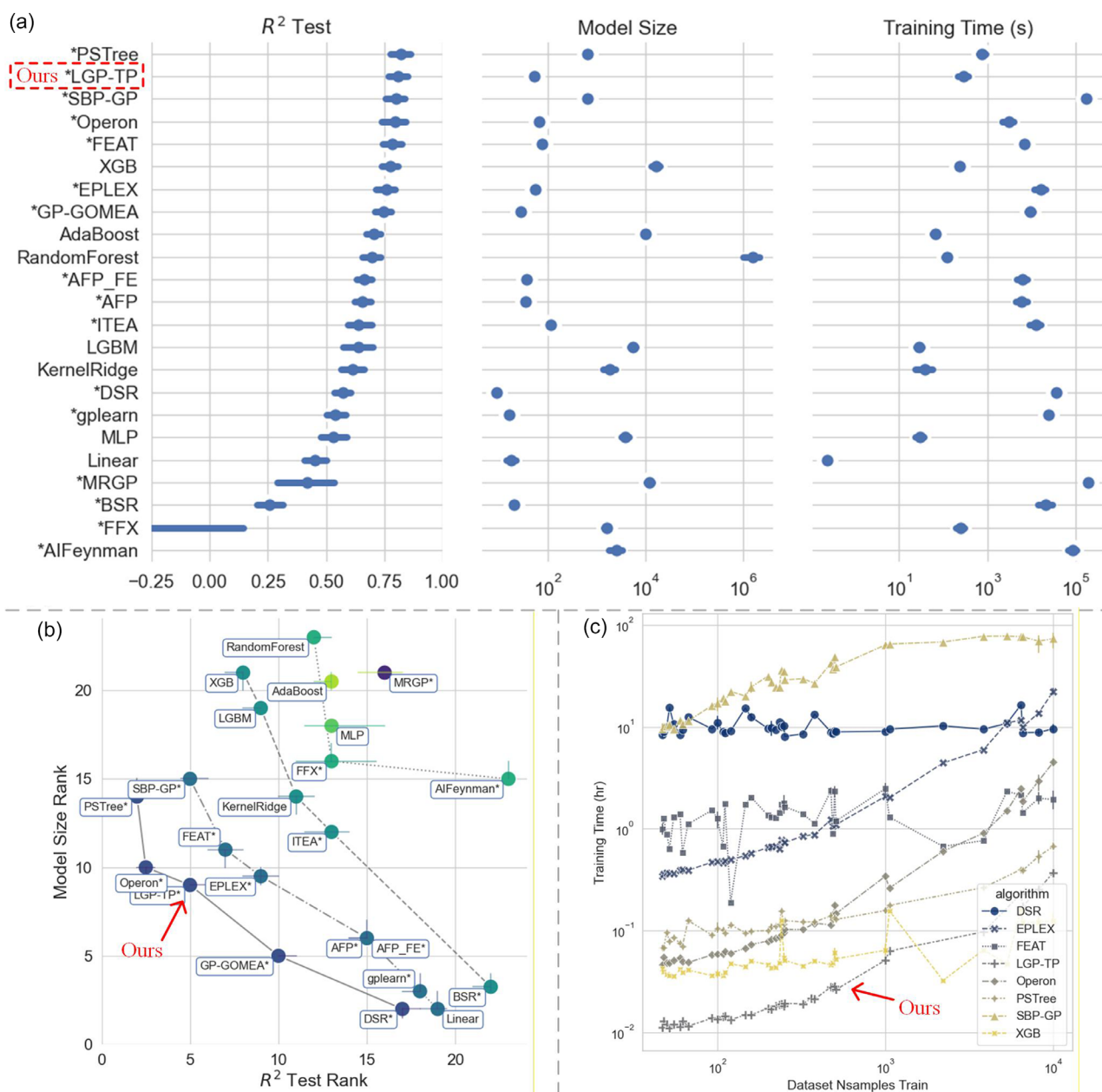


FIGURE 9 | Results of SRBench. (a) Comprehensive comparison of benchmark methods in terms of test R^2 , model size, and training time. (b) Non-dominate sets over model size rank and test R^2 rank. (c) Training time of benchmark methods over different numbers of training instances.

LGP-TP had the shortest training time in most cases, and the training time increased more slowly than many other benchmark methods. This implies good feasibility for applying LGP-TP to larger datasets.

6 | Conclusions

This paper proposes a simple and effective LGP method for analyzing fish spectral data. By introducing three kinds of tunable primitives into the evolutionary framework of basic LGP, resulting in LGP-TP, there are three major contributions.

First, LGP-TP improves the overall performance of predicting fish biomass and achieves significantly better test performance than

the compared methods, including one state-of-the-art method, on approximately half of the biomass targets. Furthermore, the synthesized regression models by LGP-TP have good interpretability. We can understand the model behaviors based on the symbolic representation, and can also highlight the informative regions of spectroscopic data. This allows human researchers to summarize new knowledge from the results. Comparison among different data treatments also shows that InGaAs Raman at 1064 nm with the wavenumber range of 580.109 to 202.533 (cm^{-1}) and SNV correction is an effective way to highlight biomass information in fish spectral data.

Second, our extension of LGP-TP to other spectral data treatments and a popular symbolic regression benchmark, SRBench, implies a promising performance of LGP-TP across a wide range of problems. The good generality allows LGP-TP to work with various

spectral data treatments in practice and synthesize concise and effective regression models for various problems.

Third, our study shows that synthesizing regression models by genetic programming methods with tunable primitives is an effective strategy to solve complex regression problems. Specifically, LGP-TP searches symbolic structures of regression models by applying the basic genetic operators of GP, which have been developed profoundly and have been shown to be effective. On the other hand, LGP-TP tunes the coefficients in the regression models (i.e., the tunable components) by approximating the target outputs based on their immediate inputs. This strategy effectively finds concise symbolic solutions with appropriate tunable parameters and avoids the cumbersome error propagation from the end of a program to its beginning.

In the future, we plan to design more effective tunable primitives for LGP and apply them to different domains. As the proposed tunable primitives in this paper only consider the linear approximation, designing tunable primitives with better learning ability is a potential direction. In addition, developing a pre-training library to selectively identify effective tunable primitives for specific tasks is a potential solution to reduce the risk of being biased to local optima.

Author Contributions

Zhixing Huang proposed the algorithm in this manuscript, implemented the experiment, and wrote the draft. **Bing Xue** and **Mengjie Zhang** helped design the algorithm. **Jeremy S. Rooney**, **Keith C. Gordon**, and **Daniel P. Killeen** collected fish spectral data and designed the experiments. All authors checked and revised the manuscript.

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Conflicts of Interest

The authors declare no conflict of interest.

Data Availability Statement

The fish spectrum data are not openly available because of commercial sensitivity. However, they are available from the data provider (Daniel.Killeen@plantandfood.co.nz) on reasonable request. The SRBench dataset is a public dataset that can be accessed via <https://cavalab.org/srbench/>.

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Appendix

A. Example Synthesized Regression Model by LGP

Figure A1 shows the amplified version of the example regression model by LGP-TP.

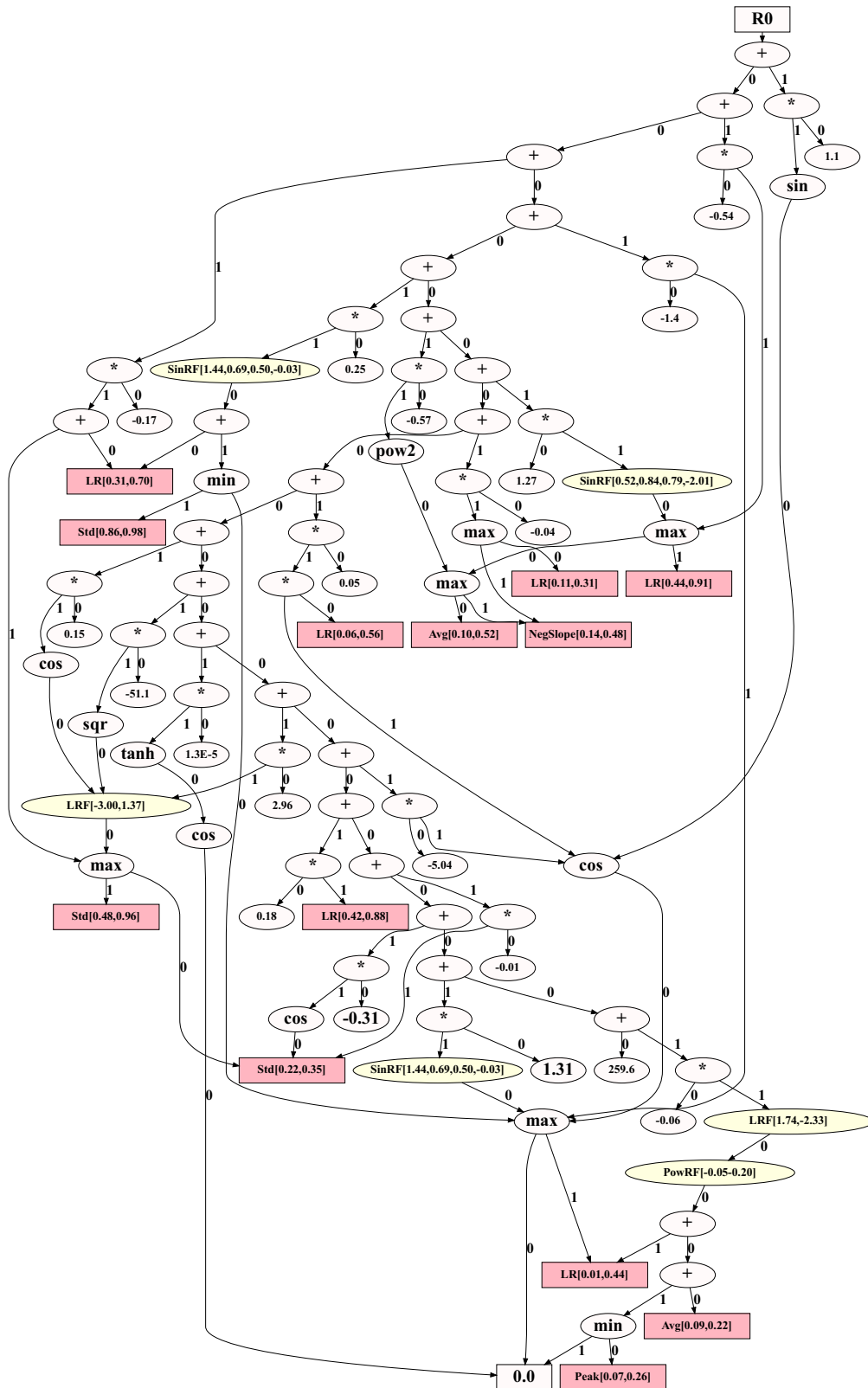


FIGURE A1 | The example synthesized regression model for predicting water in fish meat.

B. Training and Test Performance of LGP-TP

Figure B1 shows the training and test performance of LGP-TP for predicting all the ten biomass targets over its evolution.

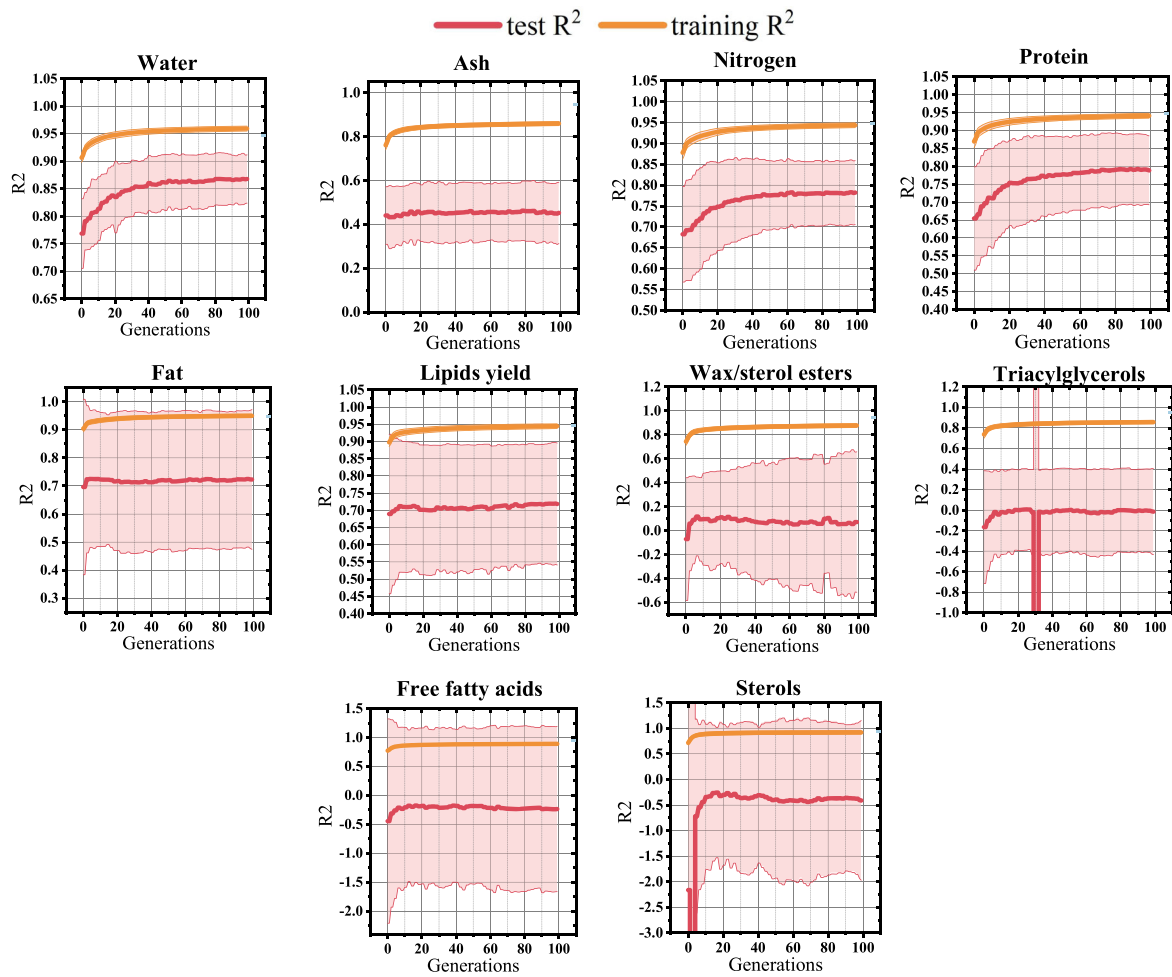


FIGURE B1 | The training and test R^2 of LGP-TP over 100 generations.